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Prognostic role of tumor-infiltrating lymphocytes assessed by hematoxylin and eosin-stained section in gastric cancer: A systematic review and meta-analysis

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Title page

Prognostic role of tumor-infiltrating lymphocytes assessed by hematoxylin and eosin-stained section in gastric cancer: A systematic review and meta-analysis

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ABSTRACT

Objective Some studies have identified tumor-infiltrating lymphocytes (TILs) in the setting of hematoxylin and eosin (H&E)-stained sections of gastric cancer, but the prognostic and clinicopathological significance of this remains unclear. The objective of this study is to evaluate the association between H&E-based TILs density and prognosis, and clinicopathological characteristics in patients with gastric cancer.

Design Systematic review and meta-analysis

Methods Databases of the Cochrane Library, PubMed, and Embase were used to find the eligible studies. Pooled hazard ratios (HRs) and relative risks (RRs) with 95% confidence intervals (95% CIs) were used to estimate the association between TILs density and overall survival (OS), and clinicopathological characteristics, respectively.

Results We enrolled 9 studies including 2835 cases for the present meta-analyses. High TILs were associated with superior OS (HR = 0.68, 95% CI: 0.52-0.87, P = 0.003) compared with low TILs. High TILs were significantly associated with lower depth of invasion (T3-4 vs T1-2) (RR = 0.58, 95% CI: 0.50-0.66, P < 0.001), less lymph node involvement (presence vs absence) (RR = 0.68, 95% CI: 0.56-0.81, P < 0.001) and earlier tumor-node-metastasis (TNM) stage (III-IV vs I-II) (RR = 0.68, 95% CI: 0.55-0.83, P < 0.001). TILs density were not associated with age, gender, Lauren classification or histological grade.

Conclusions Our meta-analysis suggests that TILs evaluated by H&E stained sections are a reliable biomarker to predict the clinical outcomes for patients with gastric cancer. Prospective observational studies are needed to further confirm our findings.

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Keywords Stomach neoplasms; Lymphocytes, Tumor-Infiltrating; Prognosis; Meta-analysis

Strengths and limitations of this study

- Compared with other methods, H&E-based TILs assessment is cost-effective, reproducible, and applicable to routine pathology practice and clinical trials.
- To our knowledge, this is the first meta-analysis to evaluate the prognostic role of TILs density evaluated by H&E stained sections in gastric cancer.
- This meta-analysis was performed by following the Preferred Reporting Items for Systematic Review and Meta-analyses Protocols.
- H&E-based TILs assessment cannot provide the information about subsets of TILs.
- Heterogeneity among different studies is inevitable.

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INTRODUCTION

Gastric cancer ranks fifth among the most frequently diagnosed carcinoma and third among the most common cause for cancer-related deaths worldwide [1]. Nearly 60% of gastric cancer patients present with late-stage cancers at diagnosis [2]. Over 40% of the patients who underwent curative resection experienced recurrence within 2 years after operation [3]. Despite improvement in the diagnosis and therapy, the prognosis of gastric cancer patients remains dismal, with 5-year overall survival (OS) rates ranging from 28% to 51% [4-7].

Among multiple clinicopathological features, such as tumor size, histologic grade, Lauren classification, lymphovascular invasion, and overexpression or amplification of the human epidermal growth factor receptor 2 (HER2), TNM staging system is regarded as a key predictor of clinical outcome in patients with gastric cancer [8]. However, heterogeneity exists in the survival outcomes of patients with

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the same TNM stage [9]. There is an unmet need to identify new biomarkers to refine the prognosis of patients. Accumulating evidence suggests tumor microenvironment (TME) (composed by fibroblast, extracellular matrix, endothelial cells, and immune cells) as a key contributor in tumorigenesis, growth, invasion, and metastasis [10]. The assessment of tumor microenvironmental features seems to have considerable prognostic value [11].

Tumor-infiltrating lymphocytes (TILs), including lymphocytes and plasma cells, reflect the host immune response to cancers, and are associated with the prognosis of various solid tumors [12-19]. The Immunoscore [14 15 20 21], which is a scoring system according to the density of $CD3^+$ and $CD8^+$ T cells at the tumor center and the invasive margin, has been demonstrated globally in stage I-III colon cancer as a superior prognostic marker than TNM staging system, lymphovascular invasion, grade of differentiation, and microsatellite instability (MSI) status [15]. High TIL levels significantly increased the sensitivity and efficacy of neoadjuvant chemotherapy in breast cancer patients [16 17], and were positively associated with survival outcomes in patients with primary cutaneous melanoma, non-small cell lung cancer, or breast cancer [17-19].

Different methods of assessing TILs are used in various studies. Compared with other methods, such as immunohistochemistry (IHC), flow cytometry, and messenger RNA profiling, hematoxylin and eosin (H&E)-based TILs assessment is cost-effective, and is applicable to clinical trials and to routine pathology practice [12 14 15]. The International Immuno-Oncology Biomarkers Working Group (IIOBWG) recommended a standardized method for TILs evaluation based on a H&E-stained section in breast cancer [22].

To date, increasing studies have evaluated the prognostic influence of TILs in gastric cancer [23 24]. However, only a few studies reported TILs assessed by H&E stained sections [25 26], and the association

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between the densities and locations of TILs and patient outcomes is unclear. The present systematic review and meta-analysis aims to evaluate the association between H&E-based TILs density and clinicopathological features, and the influence of TILs on prognosis in patients with gastric cancer.

METHODS

This study was performed based on the PRISMA statement [27]. Two independent investigators (CFT and DS) conducted literature retrieval, selected relevant studies, and extracted eligible data. Any disagreement was discussed to reach a consensus.

Search strategy

We conducted a systematic and comprehensive literature retrieval in the following databases: the Cochrane Library (1993 to February 25, 2020), PubMed (1966 to February 25, 2020), and Embase (1974 to February 25, 2020). According to MeSH terms, we used the following terms: (neoplasm, stomach or stomach neoplasm or cancer, stomach or cancer of stomach or stomach cancer or cancer of the stomach or gastric neoplasm or neoplasm, gastric or gastric cancer or cancer, gastric or gastric cancer, familial diffuse), in combination with (lymphocytes, tumor infiltrating or tumor-infiltrating lymphocytes or tumor infiltrating lymphocytes or tumor-derived activated cells or tumor derived activated cells or activated cells, tumor-derived). We also manually checked reference lists of relevant studies for any other articles.

Inclusion and exclusion criteria

The criteria for inclusion were as follows: (1) studies evaluating the correlation of TILs and either prognosis or clinicopathological characteristics in gastric cancer; (2) TILs were assessed with H&E stained sections; (3) number of patients included in each study was more than 50 cases; (4) studies with sufficient information to determine an estimate of RRs or HRs and their 95% CIs. The following studies

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were excluded: (1) duplicates, reviews, letters, comment, case reports, animal trials, cell line trials, conference abstract, meta-analysis; (2) non-English languages; (3) lacking information to calculate RRs or HRs.

Data extraction

The following information was obtained from each study: last name of the first author, publication year, country of the population, sample size, duration of follow-up, cutoff values of TILs, TILs location, number of patients with high- or low-TILs, gender, age, Lauren classification, histological grade, depth of invasion (T stage), lymph node involvement (N stage), TNM stage, HRs and 95% CIs.

Quality assessment

The criteria of Newcastle-Ottawa Scale (NOS) assessment scale [28] were used to evaluate the qualities of eligible studies. The quality score of NOS ranged from 0 to 9. Studies with a score of 6 or more were regarded as high quality.

Statistical analysis

The present meta-analysis was performed through STATA (version 12.0). The association of TILs with clinicopathological characteristics was assessed using the pooled RRs with 95% CIs. The relevant HRs and 95% CIs extracted from the original studies were combined into a pooled HR. If HRs and their 95% CIs were not reported directly, they would be calculated from Kaplan–Meier survival curves by the method previously reported by Tierney [29]. The relationship between TILs with prognosis was evaluated using the pooled HRs and 95% CIs.

The chi-squared test and I^2 test were used to assess heterogeneity between the enrolled publications. Heterogeneity was considered high for $P < 0.10$ and $I^2 > 50\%$; in such case, a random-effects model was

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used. Otherwise, a fix-effects model would be chosen. We performed subgroup analyses according to TIL locations because the function of TILs may vary across different locations. The Egger's tests and Begg's tests were conducted to detect publication bias. The source of heterogeneity and the stability of the results were assessed by sensitivity analyses.

Patient and public involvement

Patients and public were not involved in this study.

RESULTS

Literature search

A total of 1285 records were initially identified using our literature search strategy described in the materials and methods. After removing 216 duplicates, we screened 1069 titles and abstracts, and then excluded 1044 records according to the exclusion criteria. After screening the full text, we excluded another 16 studies because of insufficient data. Finally, we included 9 publications in our study, which were published from 2000 to 2019 [25 26 30-36]. The PRISMA flowchart of literature retrieval process is shown in figure 1.

Study characteristics

Among the 9 included studies, three were from South Korea [25 32 36], three from China [26 30 33], two from Japan [34 35], and one from the USA [31]. All of the studies were performed in a single-center setting. The total number of cases was 2835. All cases enrolled in this study were patients with gastric adenocarcinoma. Seven studies (n = 2613) reported data on OS [26 30-35]. Six studies [26 30 31 33-35] provided the HRs, while one study from Kim et al. [32] only provided Kaplan-Meier survival curves. The HR and its 95% CI obtained from Kaplan-Meier curves were calculated according to the methods previously

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reported [29]. The relationships between TILs density and clinicopathological characteristics were reported in 7 studies [25 26 30 32-34 36]. TILs are divided into three categories according to their locations: intratumoral TILs (iTu-TILs), stromal TILs (str-TILs), and total TILs (TILs at all the locations). Intratumoral TILs and stromal TILs were separately evaluated in 1 study [36]. Intratumoral TILs [31 35] or stromal TILs [25 33] were evaluated in 4 studies. Total TILs were reported in the other 4 studies [26 30 32 34]. Cutoff values of TILs varied across different studies. All cases were subdivided into a high-TIL group as well as a low-TIL group. The general information of the enrolled publications is summarized in table 1.

Quality assessment

According to the criteria of NOS, all 9 studies were considered to be of adequate quality for the meta-analysis (score ≥ 6 points) (table 1).

Table 1 Characteristics of the included studies

Study	Country	No. of patients (TILs-high/TILs-low)	TIL location	TNM Stage	Follow-up (month)	Cutoff value	Outcome	Quality score (NOS)
Ishigami et al. 2000 [35]	Japan	146 (69/77)	iTu-TILs	I-IV	>87	150 cells per 200 HPF	OS	6
Gochi et al. 2001 [34]	Japan	148 (37/111)	total-TILs	I-IV	>120	Moderate or marked	OS	8
Grogg et al. 2003 [31]	USA	110 (31/79)	iTu-TILs	I-IV	131	100 cells per 10 HPF	OS	8
Dai et al. 2016 [30]	China	398 (139/259)	total-TILs	I-IV	>80	total TIL score = 2	OS	7
Kang et al. 2016 [36]	South Korea	120(str-TIL 73/47; iTu-TIL 60/60)	str-TILs, iTu-TILs	I-IV	>51	Median (str-TIL 25%; iTu-TIL 30%)	NR	8
Kim et	South	102	str-TILs	EGC	NR	Median	NR	8

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al. 2017 [25]	Korea	(34/68)				(20%)		
Liu et al. 2019 [33]	China	481 (212/269)	str-TILs	I-IV	>60	Median (10%)	OS	7
Zhang et al. 2019 [26]	China	1033 (532/501)	total-TILs	I-IV	>81	total TIL score = 0.55	OS	7
Kim et al. 2019 [32]	South Korea	297 (146/151)	total-TILs	I-IV	>62	Median	OS	7

TILs, tumor-infiltrating lymphocytes; iTu-TILs, intratumoral TILs; str-TILs, stromal TILs; USA, United states of America; OS, overall survival; TNM, tumor-node-metastasis; EGC, early gastric cancer; HPF, high-power fields; NR, not report; NOS, Newcastle-Ottawa Scale.

TILs density and survival

Significant heterogeneity existed among the 7 studies including OS data (I^2 = 81.4%, P < 0.001). Accordingly, a random-effects model was used. As illustrated in figure 2, compared with low TILs density, high TILs density were significantly associated with better OS (HR = 0.68, 95% CI: 0.52-0.87, P = 0.003).

TILs density and clinicopathological characteristics

The relationships between TILs density and clinicopathological characteristics are shown in Table 2. High TILs density were significantly associated with lower depth of invasion (T3-4 vs T1-2: RR = 0.58, 95% CI: 0.50-0.66, P < 0.001), less lymph node involvement (presence vs absence: RR = 0.68, 95% CI: 0.56-0.81, P < 0.001) and earlier TNM stage (III-IV vs I-II: RR = 0.68, 95% CI: 0.55-0.83, P < 0.001). TILs density were not correlated with age, gender, Lauren classification, or histological grade. Furthermore, we carried out subgroup analyses to explore the association between TILs in different locations and OS, depth of invasion, lymph node involvement, and TNM stage.

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Table 2 The association between TILs density and clinicopathological features

Clinicopathological features	No. of studies	RR (95% CI)	P value	Heterogeneity		Effect model	Publication bias	
				I^2 (%)	P value		P_{Egger}	P_{Begg}
Age (elder vs younger)	5	0.98 (0.88 to 1.08)	0.641	0.0	0.728	Fixed	0.706	1.000
Gender (male vs female)	6	0.99 (0.90 to 1.08)	0.814	0.0	0.960	Fixed	0.200	0.133
Lauren classification (diffuse vs intestinal - mixed)	4	1.13 (0.98 to 1.30)	0.095	0.0	0.552	Fixed	0.020	0.462
Histological grade (poor vs well - moderate)	5	1.00 (0.73 to 1.37)	0.992	87.7	<0.001	Random	0.038	0.086
Depth of invasion (T3-4 vs T1-2)	4	0.58 (0.50 to 0.66)	<0.001	0.0	0.746	Fixed	0.230	0.221
Lymph node involvement (presence vs absence)	6	0.68 (0.56 to 0.81)	<0.001	63.4	0.012	Random	0.917	0.764
TNM stage (III- IV vs I-II)	4	0.68 (0.55 to 0.83)	<0.001	61.8	0.033	Random	0.797	1.000

Abbreviations: RR, risk ratio; CI, confidence interval; TNM, tumor-node-metastasis.

Subgroup analysis based on TIL locations

The results of subgroup analysis based on TIL locations are shown in figure. 3. High TILs density were significantly related to superior OS independent of their locations (iTu-TILs, P = 0.005; str-TILs, P = 0.031; total-TILs, P = 0.039) (figure 3a). Intratumoral TILs density were significantly correlated with

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lymph node involvement (P = 0.022), but not with TNM stage (P = 0.097). A borderline significant trend was shown between intratumoral TILs and invasion depth (P = 0.052). Stromal TILs density were significantly related with depth of invasion (P < 0.001), and showed borderline association with lymph node involvement (P = 0.053) and TNM stage (P = 0.054). Total TILs density were significantly associated with depth of invasion (P<0.001), lymph node involvement (P <0.001) and TNM stage (P = 0.028) (figure 3b, c, and d).

Publication bias and sensitivity analysis

Studies included in OS analysis showed no significant publication bias ($P_{Begg's}$ = 0.764, $P_{Egger's}$ = 0.851). There was no significant publication bias among studies included in clinicopathological characteristics analyses except for histological grade and Lauren classification ($P_{Egger's}$ =0.038, $P_{Egger's}$ = 0.020, respectively) (table 2).

Sensitivity analyses were performed by omitting each study in turn. The results showed that the exclusion of any single study has no significant effect on the pooled results (figure 4).

DISCUSSION

To our best knowledge, this is the first meta-analysis of the associations between TILs density evaluated by H&E stained sections and survival outcomes, and clinicopathological features in patients with gastric carcinoma. Our meta-analysis, including 2835 cases from 9 qualified studies, demonstrated that high TILs density were significantly associated with superior OS, lower depth of invasion, less lymph node involvement, and earlier TNM stage. These results indicate that TILs density evaluated by H&E stained sections can provide the important information to predict the survival outcomes of patients with gastric cancer.

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Immune checkpoint inhibitors (ICIs), which enhance host antitumor immune response by blocking checkpoint proteins [37], are novel therapeutic options for patients with advanced gastric cancer [38-40]. Oncologists and researchers show great interest in TILs because lymphocyte infiltration in tumor is a critical step for immunotherapy [41]. Patients with high TILs were more likely to achieve response to ICIs, while low TILs represent existing but partially exhausted antitumor immunity [42 43]. Our meta-analysis indicates high TILs are related to favorable clinicopathological characteristics, including lower invasion depth, less lymph node involvement, and earlier TNM stage, which is interpreted to suggest that high TILs can enhance host antitumor immunity and prevent tumor cells from invading and metastasizing. The prognostic value of TILs evaluated by H&E stained sections in gastric cancer remains controversial. In a study including 148 patients, TILs density showed no association with OS [34]. On the contrary, in another study accruing 1033 patients, high-TILs was an independent prognostic factor of OS [26]. Our meta-analysis provides strong evidence that H&E-based TIL score is a reliable prognostic biomarker in gastric cancer. This result is consistent with the previous reports across multiple cancer types [12-19 44]. Moreover, our results are in line with the meta-analyses of prognostic value of TILs evaluated by IHC in gastric cancer [45-47]. Of note, different TIL subsets detected by IHC had different prognostic values. For example, $CD3^+$ TILs and $CD8^+$ TILs had a positive effect on survival, whereas $FOXP3^+$ TILs were negatively correlated with survival [46].

IHC can define the majority of TIL subsets, provide the detailed information about the composition and functional status of TILs, and improve the evaluating accuracy of TILs [12]. However, IHC-based TIL evaluation is usually conducted on representative tissue, not on whole sections, which cannot provide comprehensive information because of the heterogeneity of TILs in different locations [12]. Moreover, the use of IHC requires additional processing and technical standardization. The implementation of TILs

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assessed by IHC sections may be difficult in low- or middle-income areas and is yet to become routine. Though H&E-based evaluations are unable to capture the complexity of TME, pathologists could be educated to score TILs on H&E-stained sections with high interpersonal concordance compared to untrained pathologists [22 42 48]. H&E-based TIL assessment is low-cost, reproducible, and applicable to routine pathology practice and clinical trials, even in resource-poor settings [12]. To date, TILs assessed by H&E-stained sections are recommended in breast cancer by IIOBWG [22]. Despite no consensus for TIL evaluation in gastric cancer, our meta-analysis suggests TILs evaluated on H&E stained sections are a reliable tool to predict survival in this cancer type.

TILs are divided into two categories based on their locations: intratumoral TILs and stromal TILs, which have been evaluated separately or together in various studies [25 26 30 32-34 36]. Lymphocytes located in tumor nests are intratumoral TILs that directly contact cancer cells, while lymphocytes dispersed in the stroma are defined as stromal TILs that do not directly interact with cancer cells [22]. Compared with intratumoral TILs, which are heterogeneously present in tumor nests and difficult to evaluate on H&E sections, stromal TILs seem to be a superior and more reproducible parameter because the density and growth pattern of tumor nests will not affect stromal TIL evaluation. In breast carcinoma, stromal TILs has been currently recommended as a principal parameter in pathological reports by IIOBWG [22]. Despite no recommendation for morphologic evaluation of TILs, our meta-analysis showed high TILs on H&E slides are associated with better OS independent of TIL locations, suggesting intratumoral TILs, stromal TILs, and total TILs might be reliable prognostic biomarkers for gastric cancer. Another meta-analysis reported similar results that high density of $CD3^+$ TILs evaluated in all areas, including cancer nest and cancer stroma, was associated with longer OS in gastric cancer [45]. However, in a study including 120 patients with Epstein-Barr virus (EBV)-positive gastric adenocarcinoma, the prognostic value was only

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seen in stromal TILs, but not in intratumoral TILs [36]. The molecular subtyping of gastric cancer suggests EBV-positive gastric cancers display gene mutation (*PIK3CA*), gene amplification (*JAK2, CD274* and *PDCD1LG2*) and DNA hypermethylation [49]. The association of TIL locations with these molecular features awaits further study. Future research that explores the biological difference of TILs in different locations is needed to better understand the host antitumor immune responses.

There are several limitations in our study. First, we calculated the HR and 95%CI from the Kaplan-Meier chart in one study [32] because of a lack of primary data from the original report; this may limit the precision of our data. Second, the methodology for evaluating TILs and the appropriate cutoff values varied across different studies. Third, most data were from East Asian populations and the prognostic value of TILs among western population should be further studied. Fourth, heterogeneity among different studies is inevitable when meta-analysis is conducted. Finally, retrospective data collection caused selection bias.

Our meta-analysis suggests that TILs evaluated by H&E stained sections are a reliable biomarker to predict prognosis for patients with gastric cancer. Prospective observational studies are needed to further confirm our findings. Future studies should focus on the biological difference between TILs in different locations.

Contributors DS and CFT designed the study, analyzed the data, drafted and edited the manuscript. HYJ and YGC performed statistical analysis, and interpreted the data, edited and wrote critical revision of the article. CXW, WBW and JPC edited and wrote critical revision of the article. All authors read and approved the final manuscript.

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Figure legends

Figure 1 PRISMA flow diagram of the literature selection process

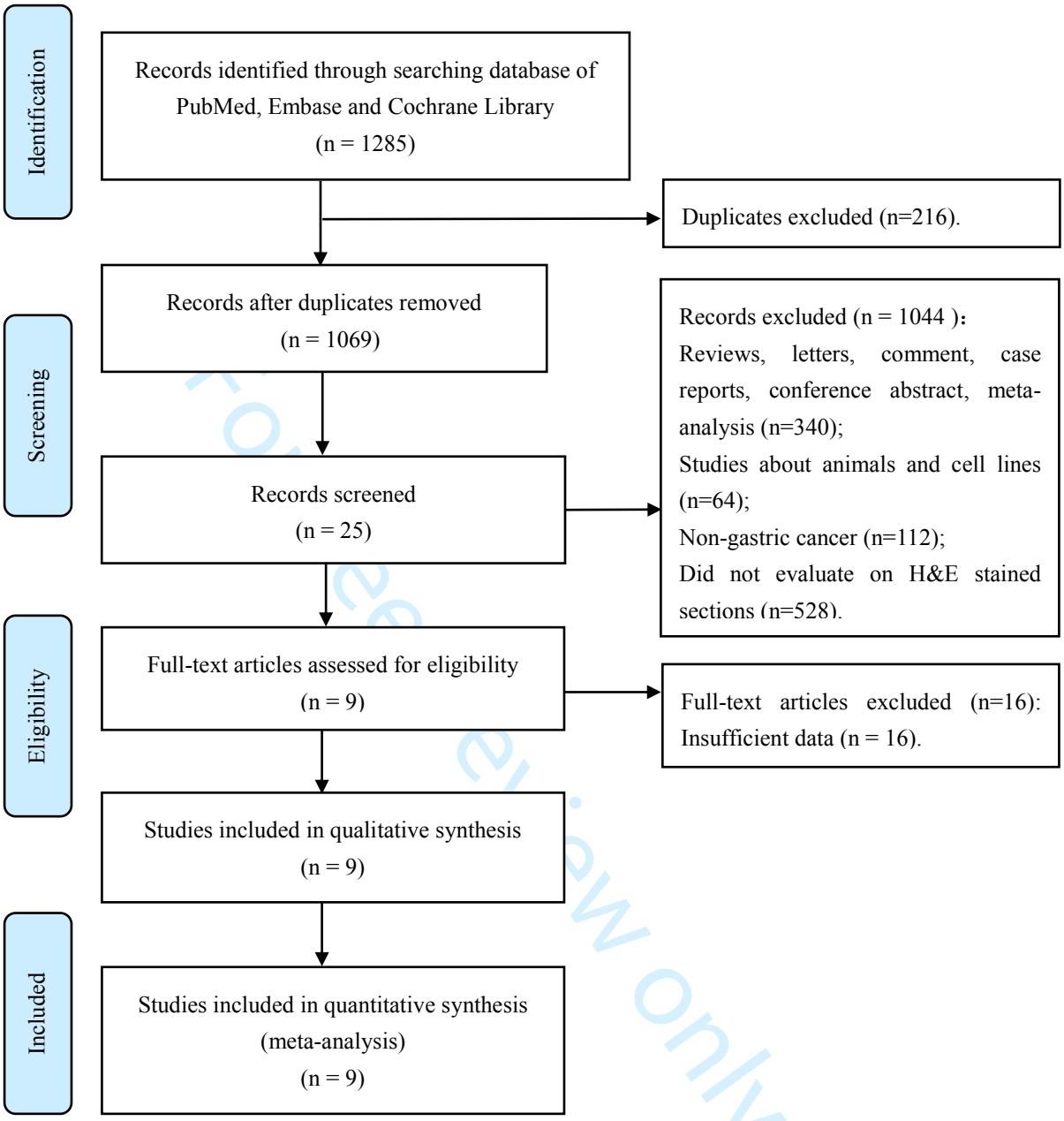
Figure 2 Forest plot for the association between TILs and overall survival of patients with gastric cancer

Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

Figure 4 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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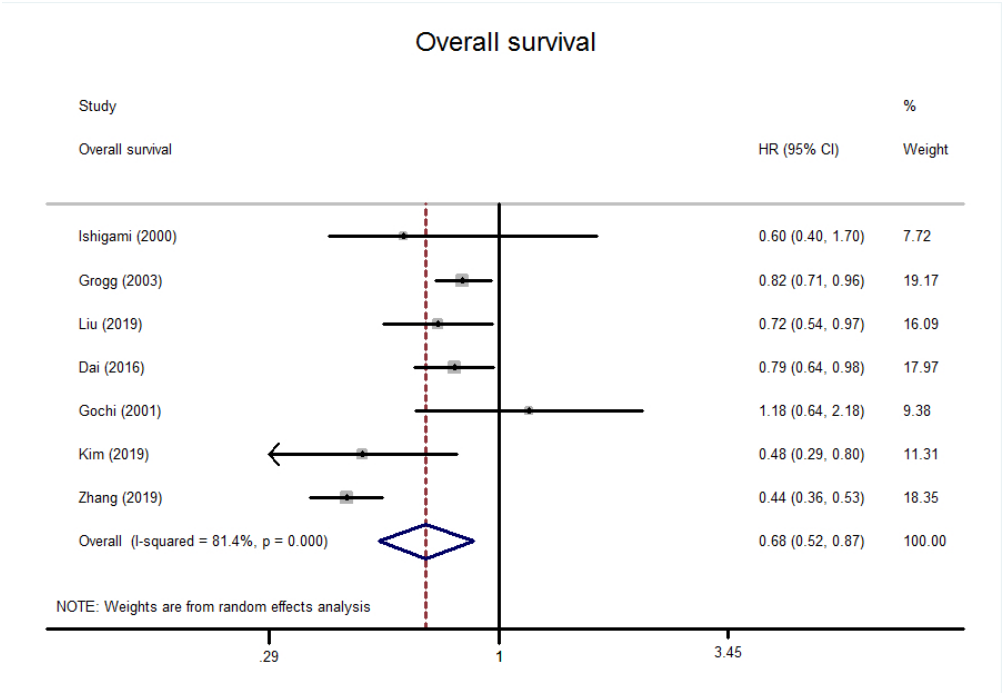


Figure 2 Forest plot for the association between TILs and overall survival of patients with gastric cancer

317x219mm (72 x 72 DPI)

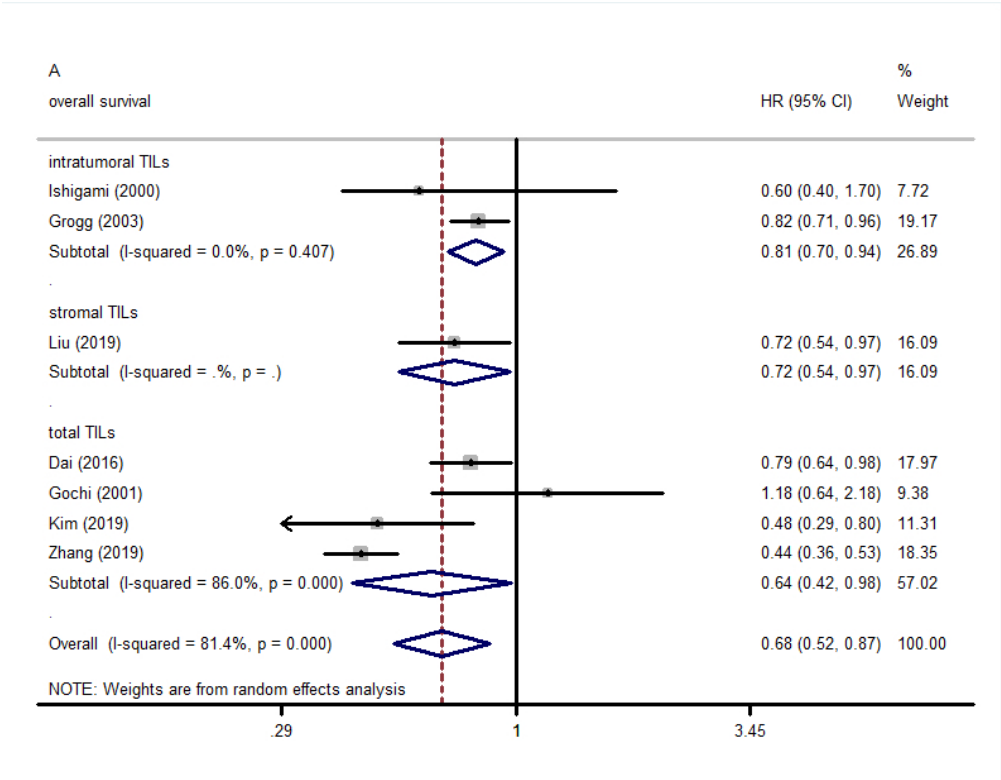


Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

279x217mm (72 x 72 DPI)

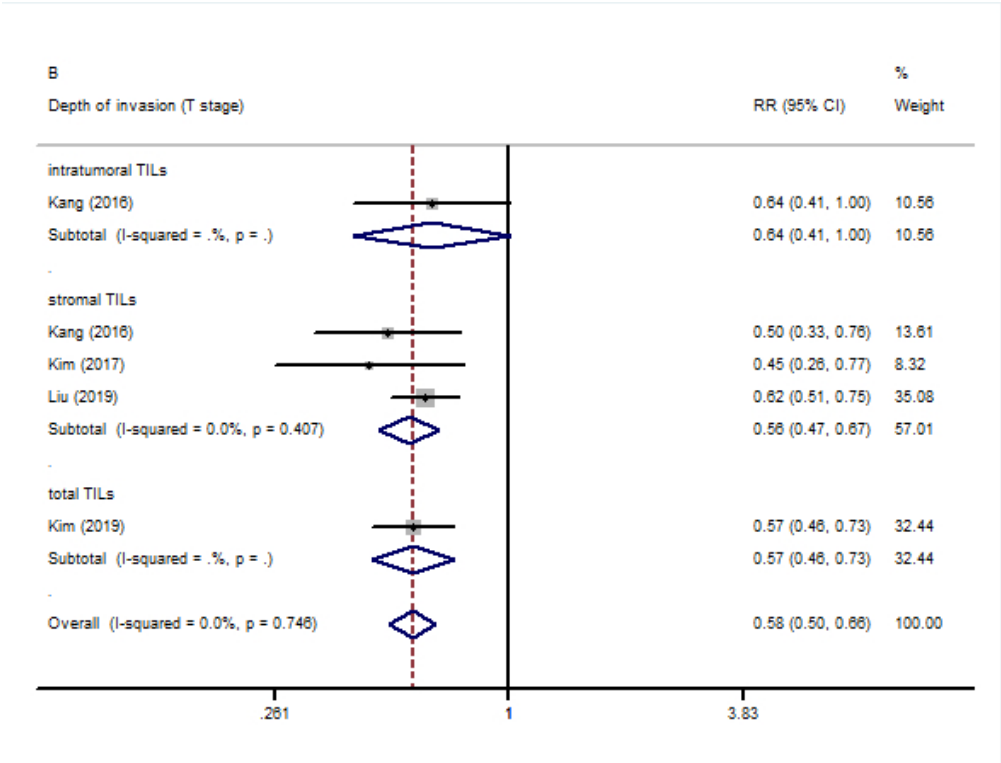


Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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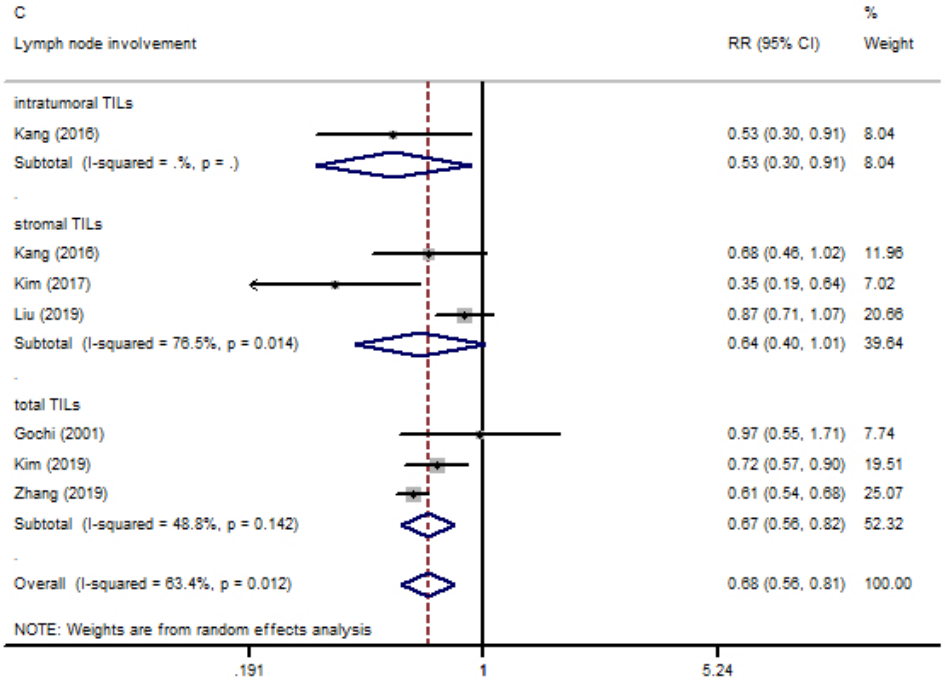


Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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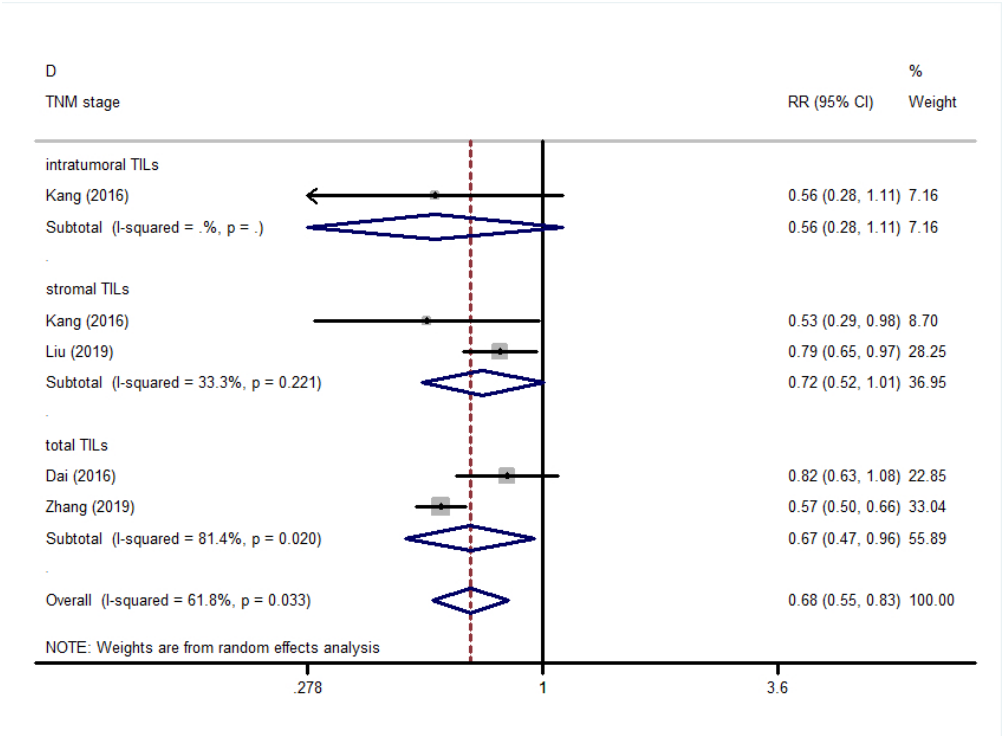


Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

298x219mm (72 x 72 DPI)

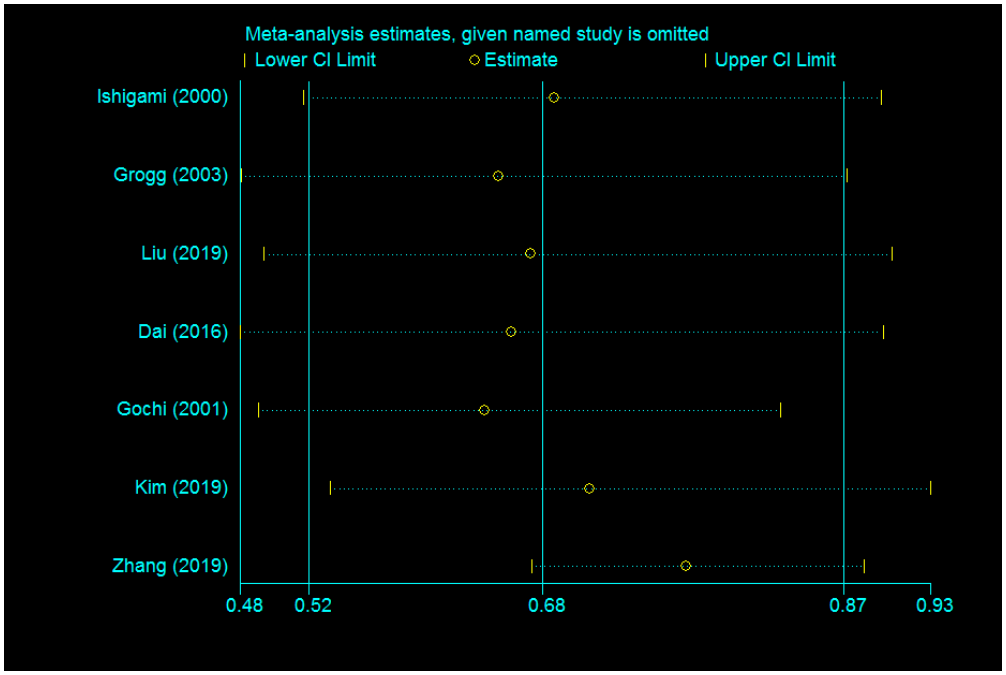


Figure 4 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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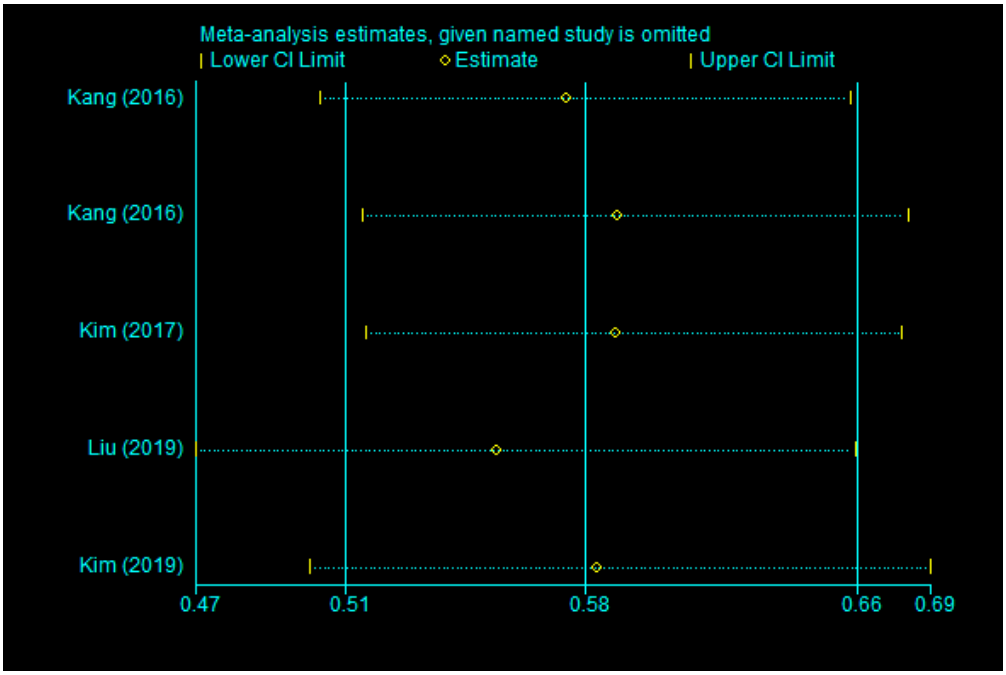


Figure 4 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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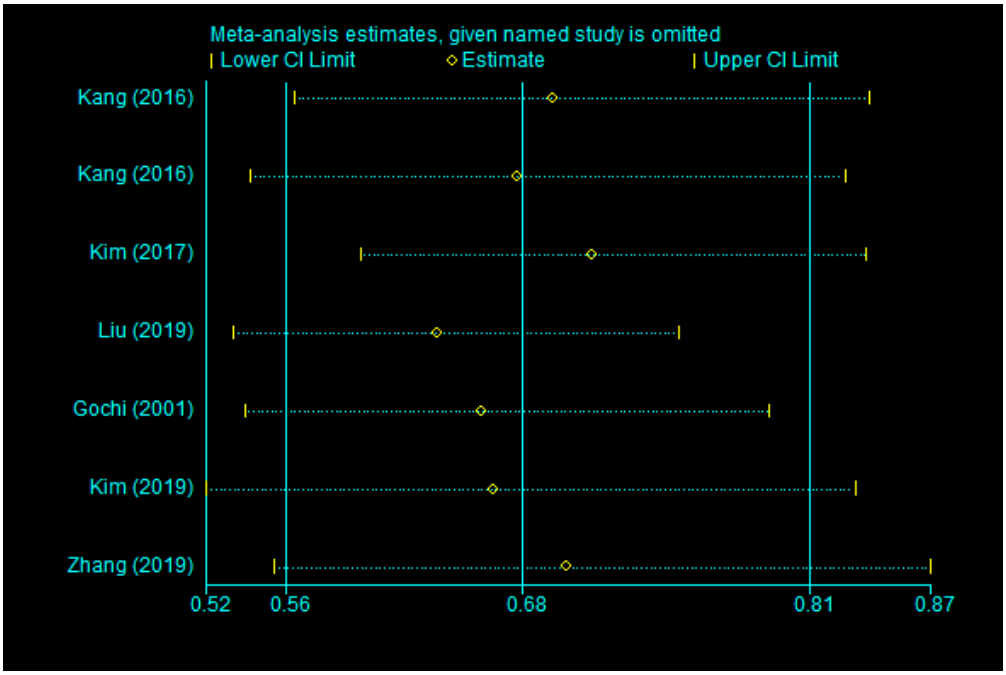


Figure 4 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

207x138mm (72 x 72 DPI)

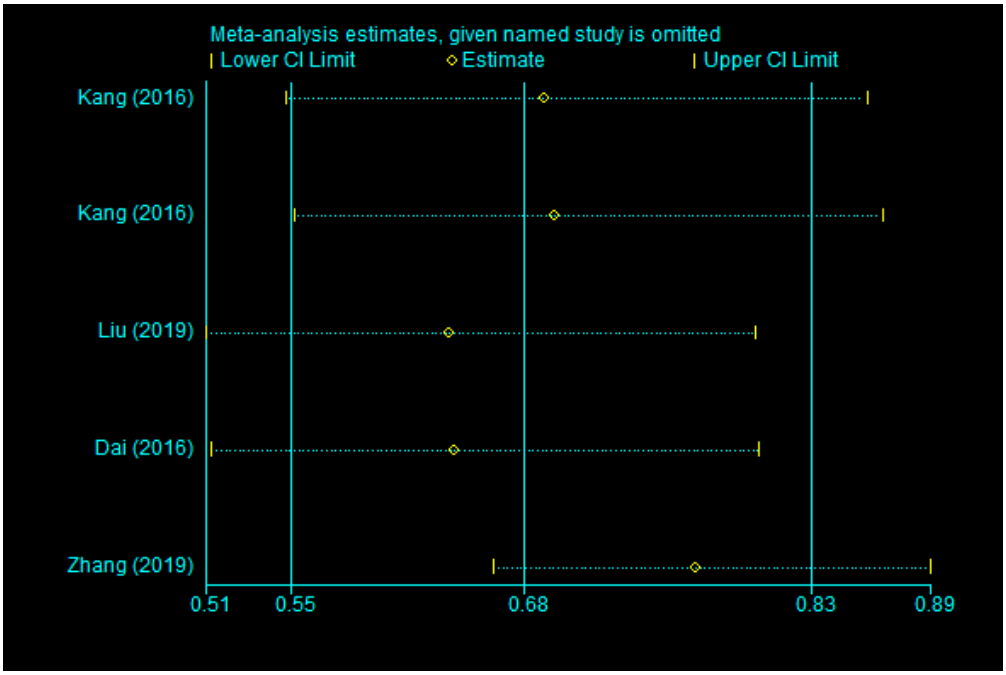


Figure 4 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9,10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9,10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11,12,13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

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Prognostic role of tumor-infiltrating lymphocytes assessed by hematoxylin and eosin-stained section in gastric cancer: A systematic review and meta-analysis

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Primary Subject Heading:	Oncology
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Title page

Prognostic role of tumor-infiltrating lymphocytes assessed by hematoxylin and eosin-stained section in gastric cancer: A systematic review and meta-analysis

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ABSTRACT

Objectives Some studies have identified tumor-infiltrating lymphocytes (TILs) in hematoxylin and eosin (H&E)-stained sections of gastric cancer, but the prognostic and clinicopathological significance of this remains unclear. The objective of this study is to evaluate the associations between H&E-based TIL density and prognosis, and clinicopathological characteristics in patients with gastric cancer.

Design Systematic review and meta-analysis

Data sources Databases of the Cochrane Library, PubMed, and Embase were searched through February 25, 2020.

Eligibility criteria Studies evaluating the correlations between TILs assessed by H&E stained sections and prognosis, and clinicopathological characteristics of gastric cancer were included

Data extraction and synthesis Relevant data were extracted and risks of bias were assessed independently by two reviewers. Hazard ratios (HRs) and relative risks (RRs) with 95% confidence intervals (95% CIs) were pooled by random-effect models to estimate the associations between TIL density and overall survival (OS), and clinicopathological characteristics, respectively.

Results We enrolled 9 studies including 2835 cases for the present meta-analysis. High TILs were associated with superior OS (HR = 0.68, 95% CI: 0.52-0.87, P = 0.003) compared with low TILs. High TILs were significantly associated with lower depth of invasion (T3-4 vs T1-2) (RR = 0.58, 95% CI: 0.50-0.66, P < 0.001), less lymph node involvement (presence vs absence) (RR = 0.68, 95% CI: 0.56-0.81, P < 0.001) and earlier tumor-node-metastasis (TNM) stage (III-IV vs I-II) (RR = 0.68, 95% CI: 0.55-0.83, P < 0.001). TIL density was not associated with age, gender, Lauren classification or histological grade. The methodology for evaluating TILs and its cutoff value varied across different studies, which might affect the results of our meta-analysis.

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Conclusions Our meta-analysis suggests that H&E-based TIL density is a reliable biomarker to predict the clinical outcomes for patients with gastric cancer. Multicenter, prospective studies are needed to further confirm our findings.

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Keywords Stomach neoplasms; Lymphocytes, Tumor-Infiltrating; Prognosis; Meta-analysis

Strengths and limitations of this study

- To our knowledge, this is the first meta-analysis to evaluate the prognostic role of TIL density evaluated by H&E stained sections in gastric cancer.
- This meta-analysis was performed by following the Preferred Reporting Items for Systematic Review and Meta-analyses Protocols.
- H&E-based TIL assessment cannot provide the information about subsets of TILs.
- Heterogeneity among different studies is inevitable.

INTRODUCTION

Gastric cancer ranks fifth among the most frequently diagnosed carcinoma and third among the most common cause for cancer-related deaths worldwide [1]. Nearly 60% of gastric cancer patients present with late-stage cancers at diagnosis [2]. Over 40% of the patients who underwent curative resection experienced recurrence within 2 years after operation [3]. Despite improvement in the diagnosis and therapy, the prognosis of gastric cancer patients remains dismal, with 5-year overall survival (OS) rates ranging from 28% to 51% [4-7].

Among multiple clinicopathological features, such as tumor size, histologic grade, Lauren classification, lymphovascular invasion, and overexpression or amplification of the human epidermal

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growth factor receptor 2 (HER2), TNM staging system is regarded as a key predictor of clinical outcome in patients with gastric cancer [8]. However, heterogeneity exists in the survival outcomes of patients with the same TNM stage [9]. There is an unmet need to identify new biomarkers to refine the prognosis of patients. Accumulating evidence suggests tumor microenvironment (TME) (composed by fibroblasts, extracellular matrix, endothelial cells, and immune cells) as a key contributor in tumorigenesis, growth, invasion, and metastasis [10]. The assessment of tumor microenvironmental features seems to have considerable prognostic value [11].

Tumor-infiltrating lymphocytes (TILs), including lymphocytes and plasma cells, reflect the host immune response to cancers, and are associated with the prognosis of various solid tumors [12-19]. The Immunoscore [14 15 20 21], which is a scoring system according to the density of $CD3^+$ and $CD8^+$ T cells at the tumor center and the invasive margin, has been demonstrated globally in stage I-III colon cancer as a superior prognostic marker than TNM staging system, lymphovascular invasion, grade of differentiation, and microsatellite instability (MSI) status [15]. High TIL levels significantly increased the sensitivity and efficacy of neoadjuvant chemotherapy in breast cancer patients [16 17], and were positively associated with survival outcomes in patients with primary cutaneous melanoma, non-small cell lung cancer, or breast cancer [17-19].

To date, there are only a few meta-analyses assessing the relationship between TILs and gastric cancer [22-24]. Most studies included in these meta-analyses evaluated TIL subsets by IHC. However, IHC-based TIL evaluation cannot provide comprehensive information because of the heterogeneity of TILs in different locations, and the use of IHC requires additional processing and technical standardization. Compared with IHC-based TIL evaluation, H&E-based TIL assessment is inexpensive, and applicable to routine pathology practice and clinical trials, even in resource-poor settings. The

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International Immuno-Oncology Biomarkers Working Group (IIOBWG) recommends a standardized method for TIL evaluation based on an H&E-stained section in breast cancer [25]. To date, there is no consensus on H&E-based TIL evaluation in gastric cancer. Our present systematic review and meta-analysis aims to evaluate the associations between H&E-based TIL density and clinicopathological features, and the influence of TILs on prognosis in patients with gastric cancer.

METHODS

This study was performed based on the PRISMA statement [26]. Two independent investigators (CFT and DS) conducted literature retrieval, selected relevant studies, and extracted eligible data. Any disagreement was discussed to reach a consensus.

Search strategy

We conducted a systematic and comprehensive literature retrieval in the following databases: the Cochrane Library (1993 to February 25, 2020), PubMed (1966 to February 25, 2020), and Embase (1974 to February 25, 2020). According to MeSH terms, we used the following terms: (neoplasm, stomach or stomach neoplasm or cancer, stomach or cancer of stomach or stomach cancer or cancer of the stomach or gastric neoplasm or neoplasm, gastric or gastric cancer or cancer, gastric or gastric cancer, familial diffuse), in combination with (lymphocytes, tumor infiltrating or tumor-infiltrating lymphocytes or tumor infiltrating lymphocytes or tumor-derived activated cells or tumor derived activated cells or activated cells, tumor-derived). The full electronic search strategy for PubMed can be accessed in the Supplementary File (supplementary tables S1-S2). We also manually checked reference lists of relevant studies for any other articles.

Inclusion and exclusion criteria

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The criteria for inclusion were as follows: (1) studies evaluating the correlation of TILs and either prognosis or clinicopathological characteristics in gastric cancer; (2) TILs were assessed with H&E stained sections; (3) number of patients included in each study was more than 50 cases; (4) studies with sufficient information to determine an estimate of RRs or HRs and their 95% CIs. The following studies were excluded: (1) duplicates, reviews, letters, comments, case reports, animal trials, cell line trials, conference abstracts, meta-analyses; (2) non-English languages; (3) lacking information to calculate RRs or HRs.

Data extraction

The following information was obtained from each study: last name of the first author, publication year, country, sample size, duration of follow-up, cutoff values of TILs, TIL location, number of patients with high- or low-TILs, gender, age, Lauren classification, histological grade, depth of invasion (T stage), lymph node involvement (N stage), TNM stage, HRs and 95% CIs.

Quality assessment

The criteria of Newcastle-Ottawa Scale (NOS) assessment scale [27] were used to evaluate the qualities of eligible studies. The quality score of NOS ranged from 0 to 9. Studies with a score of 6 or more were regarded as high quality.

Statistical analysis

The present meta-analysis was performed through STATA (version 12.0). The associations of TILs with clinicopathological characteristics were assessed using the pooled RRs with 95% CIs. The relevant HRs and 95% CIs extracted from the original studies were combined into a pooled HR. If HRs and their 95% CIs were not reported directly, they would be calculated from Kaplan–Meier survival curves by the

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method previously reported by Tierney [28]. The relationship between TILs with prognosis was evaluated using the pooled HRs and 95% CIs.

We calculated all effects on the basis of random-effect models. The chi-squared test and I^2 test were used to assess heterogeneity between the enrolled publications. Heterogeneity was considered high for $P < 0.15$ and $I^2 > 50\%$ [29]. We performed subgroup analyses according to TIL locations because the function of TILs may vary across different locations. The Egger's tests [30] and Begg's tests [31] were conducted to detect publication bias. The trim and fill method was used to evaluate the effects of publication bias on the pooled results [32]. The source of heterogeneity and the stability of the results were assessed by sensitivity analyses.

Patient and public involvement

There was no patient or public involvement in the design or conduct of this study.

RESULTS

Literature search

A total of 1285 records were initially identified using our literature search strategy described in the materials and methods. After removing 216 duplicates, we screened 1069 titles and abstracts, and then excluded 1044 records according to the exclusion criteria. After screening the full text, we excluded another 16 studies because of insufficient data. Finally, we included 9 publications in our study, which were published from 2000 to 2019 [33-41]. The PRISMA flowchart of literature retrieval process is shown in figure 1.

Study characteristics

Among the nine observational studies included in this meta-analysis, three were from South Korea [33-35],

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three from China [36-38], two from Japan [39 40], and one from the USA [41]. All of the studies were performed in a single-center setting. The total number of cases was 2835. All cases enrolled in this study were patients with gastric adenocarcinoma. Seven studies (n = 2613) reported data on OS [34 36-41]. Six studies [36-41] provided the HRs, while one study from Kim et al. [34] only provided Kaplan-Meier survival curves. The HR and its 95% CI obtained from Kaplan-Meier curves were calculated according to the methods previously reported [28]. The relationships between TIL density and clinicopathological characteristics were reported in 7 studies [33-39]. TILs are divided into three categories according to their locations: intratumoral TILs (iTu-TILs), stromal TILs (str-TILs), and total TILs (TILs at all the locations). Intratumoral TILs and stromal TILs were separately evaluated in 1 study [35]. Intratumoral TILs [40 41] or stromal TILs [33 38] were evaluated in 4 studies. Total TILs were reported in the other 4 studies [34 36 37 39]. TIL density was evaluated manually in 8 [33-39 41] of the 9 included studies. The remaining study was evaluated digitally [40]. Four studies [33 35 36 38] evaluated TILs based on the IIOBWG recommendations, and other five studies [34 37 39-41] evaluated TILs based on different approaches. Cutoff values of TILs varied across different studies. All cases were subdivided into a high-TIL group as well as a low-TIL group. The general information of the enrolled publications is summarized in table 1.

Quality assessment

According to the criteria of NOS, all 9 studies were considered to be of adequate quality for the meta-analysis (score ≥ 6 points) (table 1).

Study	Country	No. of patients (TILs-high/TILs-low)	TIL location	TNM Stage	Follow-up (month)	Cutoff value	Outcome	Quality score (NOS)

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Ishigami et al. 2000 [40]	Japan	146 (69/77)	iTu-TILs	I-IV	>87	150 cells per 200 HPF	OS	6
Gochi et al. 2001 [39]	Japan	148 (37/111)	total-TILs	I-IV	>120	Moderate or marked	OS	8
Grogg et al. 2003 [41]	USA	110 (31/79)	iTu-TILs	I-IV	131	100 cells per 10 HPF	OS	8
Dai et al. 2016 [37]	China	398 (139/259)	total-TILs	I-IV	>80	total TIL score = 2	OS	7
Kang et al. 2016 [35]	South Korea	120(str-TILs 73/47; iTu-TILs 60/60)	str-TILs, iTu-TILs	I-IV	>51	Median (str-TILs 25%; iTu-TILs 30%)	NR	8
Kim et al. 2017 [33]	South Korea	102 (34/68)	str-TILs	EGC	NR	Median (20%)	NR	8
Liu et al. 2019 [38]	China	481 (212/269)	str-TILs	I-IV	>60	Median (10%)	OS	7
Zhang et al. 2019 [36]	China	1033 (532/501)	total-TILs	I-IV	>81	total TIL score = 0.55	OS	7
Kim et al. 2019 [34]	South Korea	297 (146/151)	total-TILs	I-IV	>62	Median	OS	7

TILs, tumor-infiltrating lymphocytes; iTu-TILs, intratumoral TILs; str-TILs, stromal TILs; USA, United states of America; OS, overall survival; TNM, tumor-node-metastasis; EGC, early gastric cancer; HPF, high-power fields; NR, not report; NOS, Newcastle-Ottawa Scale.

TIL density and survival

Significant heterogeneity existed among the 7 studies including OS data (I^2 = 81.4%, P < 0.001). Accordingly, a random-effect model was used. As illustrated in figure 2, compared with low TIL density, high TIL density was significantly associated with better OS (HR = 0.68, 95% CI: 0.52-0.87, P = 0.003).

TIL density and clinicopathological characteristics

The relationships between TIL density and clinicopathological characteristics are shown in Table 2. High

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TIL density was significantly associated with lower depth of invasion (T3-4 vs T1-2: RR = 0.58, 95% CI: 0.50-0.66, P < 0.001), less lymph node involvement (presence vs absence: RR = 0.68, 95% CI: 0.56-0.81, P < 0.001) and earlier TNM stage (III-IV vs I-II: RR = 0.68, 95% CI: 0.55-0.83, P < 0.001). TIL density was not correlated with age, gender, Lauren classification, or histological grade. Furthermore, we carried out subgroup analyses to explore the associations between TILs in different locations and OS, depth of invasion, lymph node involvement, and TNM stage.

Table 2 The association between TIL density and clinicopathological features

				Heterogeneity			Publication bias	
Clinicopathological features	**No. of studies**	**RR (95% CI)**	**P value**	I^2 (%)	P value	**Effect model**	P_{Egger}	P_{Begg}
Age (elder vs younger)	5	0.97 (0.88 to 1.08)	0.617	0.0	0.728	Random	0.706	1.000
Gender (male vs female)	6	0.99 (0.90 to 1.08)	0.750	0.0	0.960	Random	0.200	0.133
Lauren classification (diffuse vs intestinal - mixed)	4	1.13 (0.98 to 1.30)	0.097	0.0	0.552	Random	0.020	0.462
Histological grade (poor vs well - moderate)	5	1.00 (0.73 to 1.37)	0.992	87.7	<0.001	Random	0.038	0.086
Depth of invasion (T3-4 vs T1-2)	4	0.58 (0.51 to 0.66)	<0.001	0.0	0.746	Random	0.230	0.221
Lymph node involvement (presence vs absence)	6	0.68 (0.56 to 0.81)	<0.001	63.4	0.012	Random	0.917	0.764
TNM stage (III- IV vs	4	0.68	<0.001	61.8	0.033	Random	0.797	1.000

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I-II)	(0.55 to 0.83)	

Abbreviations: RR, risk ratio; CI, confidence interval; TNM, tumor-node-metastasis.

Subgroup analysis based on TIL locations

The results of subgroup analysis based on TIL locations are shown in figure. 3. High TIL density was significantly related to superior OS independent of their locations (iTu-TILs, P = 0.005; str-TILs, P = 0.031; total-TILs, P = 0.039) (figure 3a). Intratumoral TIL density was significantly correlated with lymph node involvement (P = 0.022), but not with TNM stage (P = 0.097). A borderline significant trend was shown between intratumoral TILs and invasion depth (P = 0.052). Stromal TIL density was significantly related with depth of invasion (P < 0.001), and showed borderline association with lymph node involvement (P = 0.053) and TNM stage (P = 0.054). Total TIL density was significantly associated with depth of invasion (P<0.001), lymph node involvement (P <0.001) and TNM stage (P = 0.028) (figure 3b, c, and d).

Publication bias and sensitivity analysis

Studies included in OS analysis showed no significant publication bias ($P_{Begg's}$ = 0.764, $P_{Egger's}$ = 0.851). There was no significant publication bias among studies included in clinicopathological characteristics analyses except for histological grade and Lauren classification ($P_{Egger's}$ =0.038, $P_{Egger's}$ = 0.020, respectively) (table 2). The trim and fill method indicated that studies were unlikely to be missing in the analysis of the association between TILs and histological grade, while two studies were found missing in the analysis of the association between TILs and Lauren classification (figure 4). The effects of TILs in histological grade and Lauren classification were stable and not changed after having adjusted the publication bias.

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Sensitivity analyses were performed by omitting each study in turn. The results showed that the exclusion of any single study has no significant effect on the pooled results (figure 5).

DISCUSSION

To our best knowledge, this is the first meta-analysis of the associations between TIL density evaluated by H&E stained sections and survival outcomes, and clinicopathological features in patients with gastric carcinoma. Our meta-analysis, including 2835 cases from 9 qualified observational studies, demonstrated that high TIL density was significantly associated with superior OS, lower depth of invasion, less lymph node involvement, and earlier TNM stage. These results indicate that TIL density evaluated by H&E stained sections can provide important information to predict the survival outcomes of patients with gastric cancer.

Immune checkpoint inhibitors (ICIs), which enhance host antitumor immune response by blocking checkpoint proteins [42], are novel therapeutic options for patients with advanced gastric cancer [43-45]. Oncologists and researchers show great interest in TILs because lymphocyte infiltration in tumor is a critical step for immunotherapy [46]. Our previous study in colon cancer (n=1532) indicated tumors with high TILs had distinct tumor biology, were more likely to present with a lower T and N stage, well or moderate histologic grade, right-sided primary location, mutated *BRAF* ^{V600E}, and dMMR [47]. Patients with high TILs were more likely to achieve response to ICIs, while low TILs represent existing but partially exhausted antitumor immunity [48 49]. High TILs were also associated with the efficacy of chemotherapy in gastric cancer. Li et al. [50] found that high $CD3^+$ or $FoxP3^+$ T-cell infiltration was associated with better OS in patients with stage II–III gastric cancer treated with FOLFOX. Our meta-analysis indicates high TILs are related to favorable clinicopathological characteristics, including lower invasion depth, less lymph node involvement, and earlier TNM stage, which is interpreted to suggest that

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high TILs can enhance host antitumor immunity and prevent tumor cells from invading and metastasizing. The prognostic value of TILs evaluated by H&E stained sections in gastric cancer remains controversial. In a study including 148 patients, TIL density showed no association with OS [39]. On the contrary, in another study accruing 1033 patients, high-TIL density was an independent prognostic factor of OS [36]. Our meta-analysis provides strong evidence that H&E-based TIL density is a reliable prognostic biomarker in gastric cancer. This result is consistent with previous reports across multiple cancer types [12-19 47]. Moreover, our results are in line with the meta-analyses of prognostic value of TILs evaluated by IHC in gastric cancer [22-24]. Of note, different TIL subsets detected by IHC had different prognostic values. For example, $CD3^+$ TILs and $CD8^+$ TILs had positive effects on survival, whereas $FOXP3^+$ TILs were negatively correlated with survival [23].

IHC can define the majority of TIL subsets, provide the detailed information about the composition and functional status of TILs, and improve the evaluating accuracy of TILs [12]. However, IHC-based TIL evaluation is usually conducted on representative tissues, not on whole sections, which cannot provide comprehensive information because of the heterogeneity of TILs in different locations [12]. Moreover, the use of IHC requires additional processing and technical standardization. The implementation of TILs assessed by IHC sections may be difficult in low- or middle-income areas and is yet to become routine. Though H&E-based evaluations are unable to capture the complexity of TME, pathologists could be educated to score TILs on H&E-stained sections with high interpersonal concordance compared to untrained pathologists [25 48 51]. H&E-based TIL assessment is inexpensive, and applicable to routine pathology practice and clinical trials, even in resource-poor settings [12]. To date, TILs assessed by H&E-stained sections are recommended in breast cancer by IIOBWG [25]. Despite no consensus for TIL evaluation and its cutoff in gastric cancer, our meta-analysis suggests TIL density evaluated on H&E

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stained sections is a reliable tool to predict survival in this cancer type. Future study is needed to explore the optimal evaluation method and cutoff value for TILs.

TILs are divided into two categories based on their locations: intratumoral TILs and stromal TILs, which have been evaluated separately or together in various studies [33-41]. Lymphocytes located in tumor nests are intratumoral TILs that directly contact cancer cells, while lymphocytes dispersed in the stroma are defined as stromal TILs that do not directly interact with cancer cells [22]. Stromal TILs have been currently recommended as a principal parameter in breast carcinoma pathological reports by IIOBWG [25]. To date, prognostic value of TIL's locations vary in different reports and there are no recommendations for evaluation of TILs in gastric cancer. Our meta-analysis showed high TILs on H&E slides were associated with better OS independent of TIL locations, suggesting intratumoral TILs, stromal TILs, and total TILs might be reliable prognostic biomarkers for gastric cancer. A meta-analysis reported that high density of $CD3^+$ TILs evaluated in all areas, including cancer nest and cancer stroma, was associated with longer OS in gastric cancer [22]. One study indicated high intratumoral TILs were related to prolonged OS in patients with gastric adenocarcinoma [41]. On the contrary, in another study including 120 patients with Epstein-Barr virus (EBV)-positive gastric adenocarcinoma, the prognostic value was only seen in stromal TILs, but not in intratumoral TILs [35]. EBV-positive gastric cancers displayed gene mutation (*PIK3CA*), gene amplification (*JAK2, CD274* and *PDCD1LG2*) and DNA hypermethylation [52]. The association of TIL locations with molecular features awaits further study. Future research that explores the biological difference of TILs in different locations is needed to better understand the host antitumor immune responses.

There are several limitations in our study. First, we calculated the HR and 95%CI from the Kaplan-Meier chart in one study [34] because of a lack of primary data from the original report; this may limit the

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precision of our data. Second, the methodology for evaluating TILs and its cutoff value varied across different studies, which might affect the results of our meta-analysis. Third, most data were from East Asian populations and the prognostic value of TILs among western populations should be further studied. Fourth, heterogeneity among different studies is inevitable when meta-analysis is conducted. In addition, publication bias in histological grades and Lauren classification was detected and adjusted by the trim and fill method in our meta-analysis. Finally, retrospective data collection caused selection bias.

Our meta-analysis suggests that TIL density evaluated by H&E stained sections in observational studies is a reliable biomarker to predict prognosis in patients with gastric cancer. Multicenter, prospective studies are needed to determine the optimal evaluation method (IHC or H&E stained sections or others) and cutoff values for TILs. Future studies should also focus on the biological difference between TILs in different locations.

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Contributors DS and CFT designed the study, analyzed the data, drafted and edited the manuscript. HYJ and YGC performed statistical analysis, and interpreted the data, edited and wrote critical revision of the article. CXW, WBW and JPC edited and wrote critical revision of the article. All authors read and approved the final manuscript.

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supplementary information.

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Figure legends

Figure 1 PRISMA flow diagram of the literature selection process

Figure 2 Forest plot for the association between TILs and overall survival of patients with gastric cancer

Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

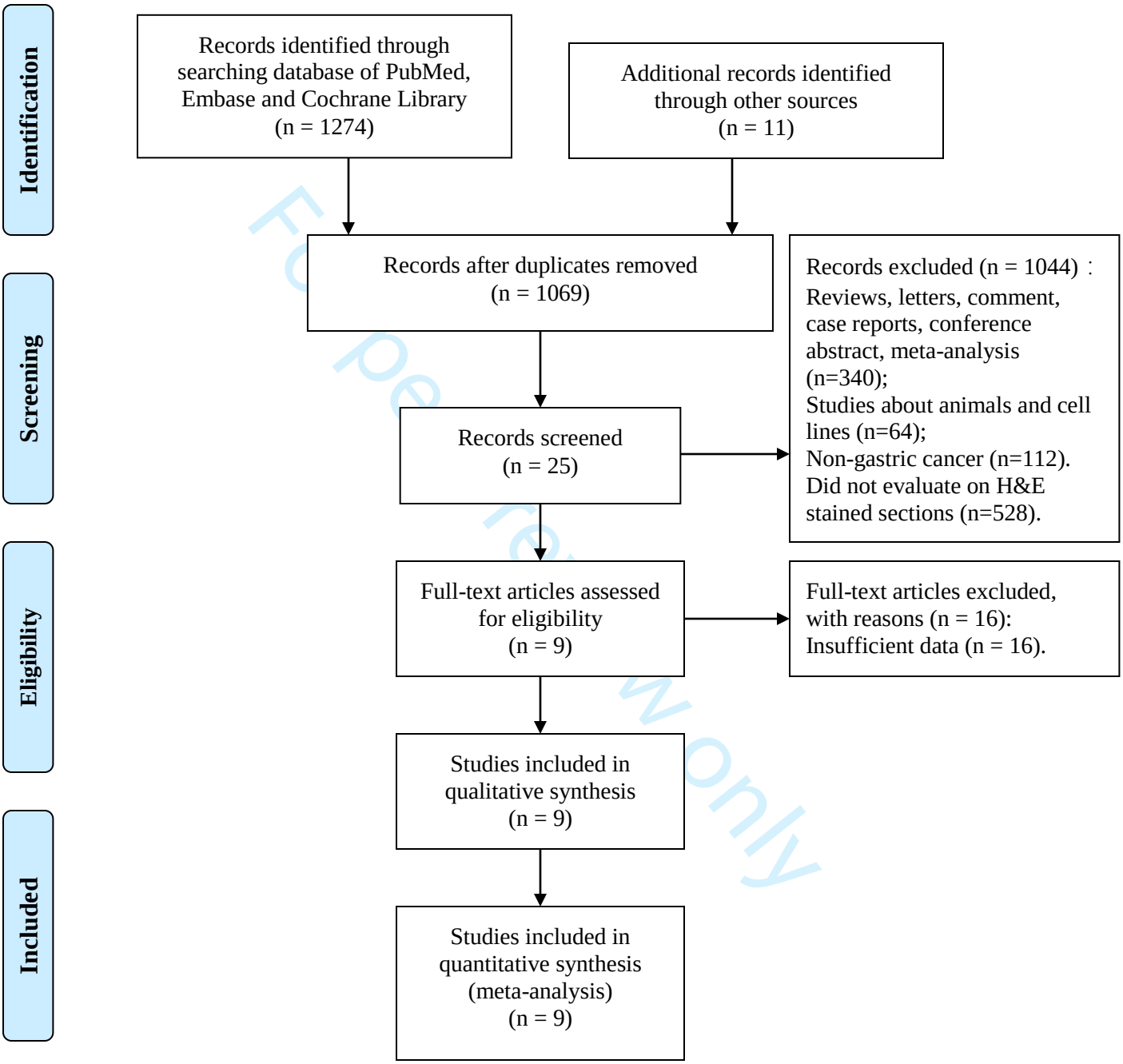
Figure 4 Trim and fill analysis of studies included in this meta-analysis. (A) Histological grade. (B) Lauren classification.

Figure 5 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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PRISMA 2009 Flow Diagram



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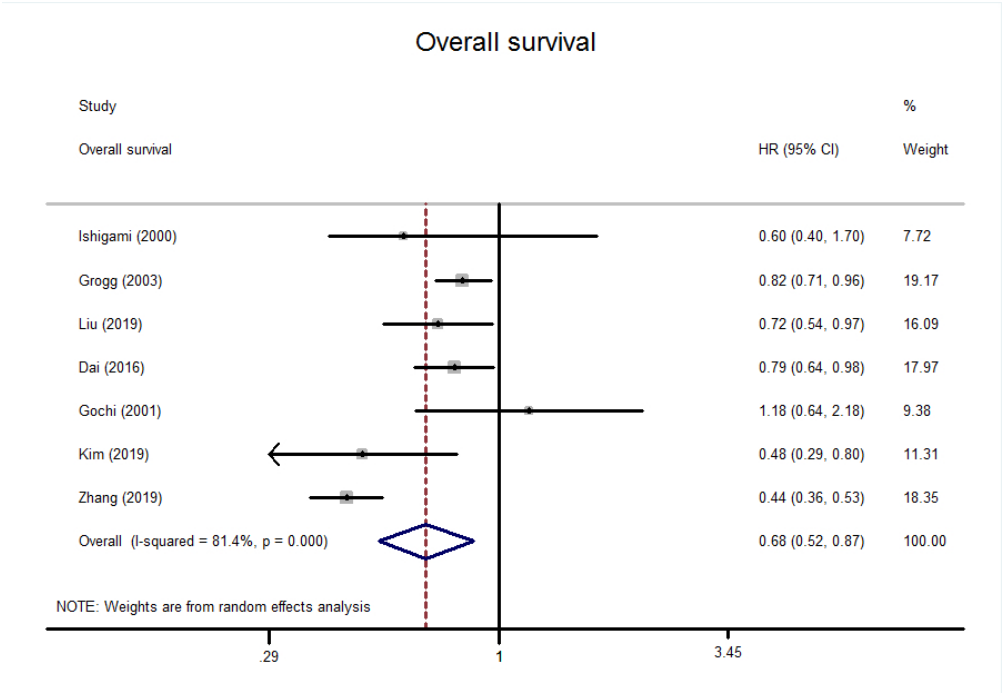


Figure 2 Forest plot for the association between TILs and overall survival of patients with gastric cancer

317x219mm (72 x 72 DPI)

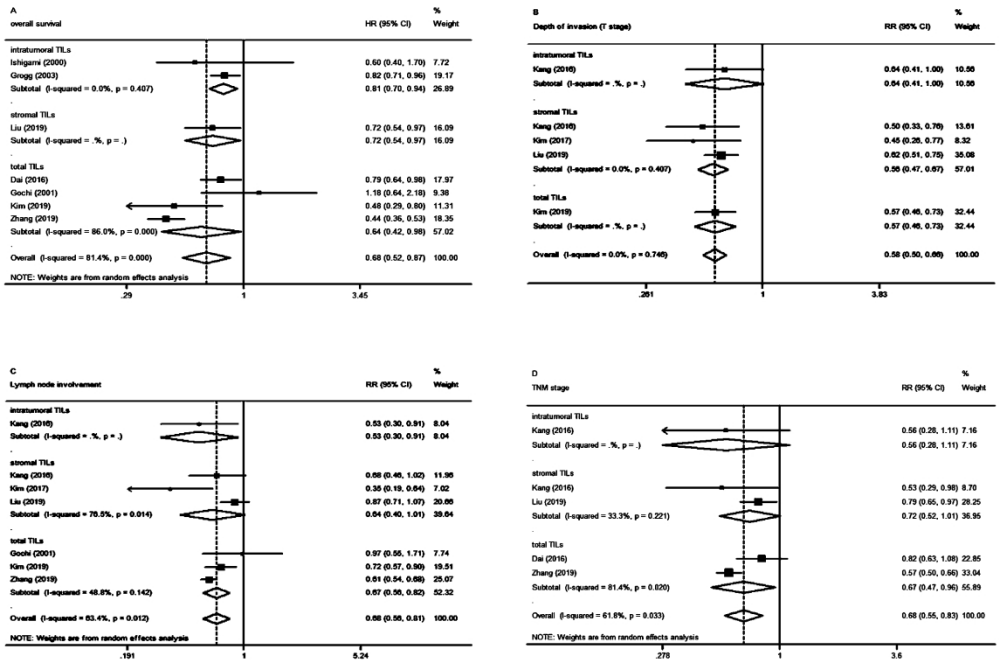


Figure 3 Forest plots of the subgroup analysis for the association between TILs in different locations and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

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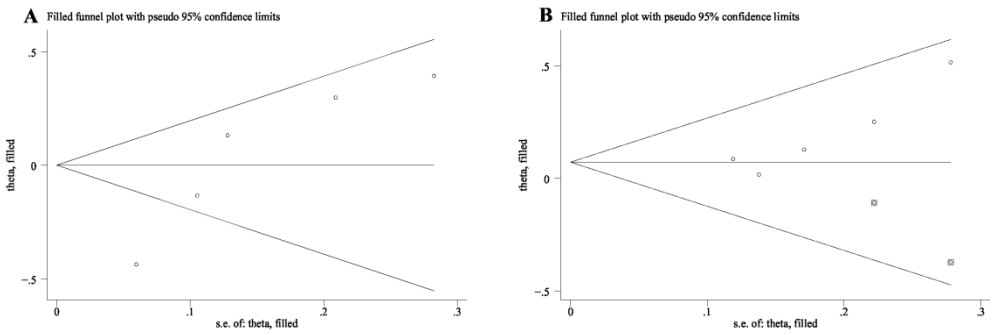


Figure 4 Trim and fill analysis of studies included in this meta-analysis. (A) Histological grade. (B) Lauren classification.

204x70mm (300 x 300 DPI)

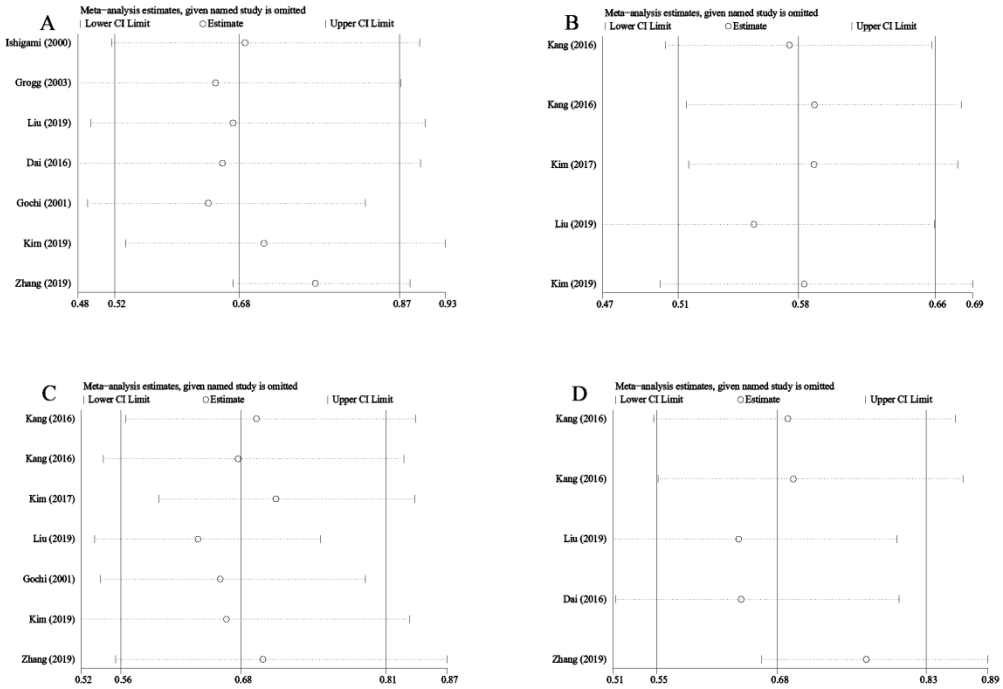


Figure 5 Sensitivity analysis of the association between TILs and prognosis, and clincopathological features. (A) Overall survival. (B) Depth of invasion. (C) Lymph node involvement. (D) TNM stage.

193x132mm (300 x 300 DPI)

Supplementary Table S1: The electronic search strategy for PubMed

Search terms
1. "Stomach Neoplasms"[MeSH]
2. ((((((((((((((((((Neoplasm, Stomach) OR Stomach Neoplasm) OR Neoplasms, Stomach) OR Gastric Neoplasms) OR Gastric Neoplasm) OR Neoplasm, Gastric) OR Neoplasms, Gastric) OR Cancer of Stomach) OR Stomach Cancers) OR Gastric Cancer) OR Cancer, Gastric) OR Cancers, Gastric) OR Gastric Cancers) OR Stomach Cancer) OR Cancer, Stomach) OR Cancers, Stomach) OR Cancer of the Stomach) OR Gastric Cancer, Familial Diffuse)
3. 1 or 2
4. "lymphocytes, tumor infiltrating"[MeSH]
5. ((((((((((Lymphocytes, Tumor Infiltrating) OR Tumor-Infiltrating Lymphocytes) OR Lymphocyte, Tumor-Infiltrating) OR Tumor Infiltrating Lymphocytes) OR Tumor-Infiltrating Lymphocyte) OR Tumor-Derived Activated Cells) OR Activated Cell, Tumor-Derived) OR Activated Cells, Tumor-Derived) OR Tumor Derived Activated Cells) OR Tumor-Derived Activated Cell))
6. 4 or 5
7. 3 and 6

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Supplementary Table S2: Electronic links to full search strategy in PubMed

https://pubmed.ncbi.nlm.nih.gov/?sort=date&term=%22Stomach+Neoplasms%22%5BMesh%5D
https://pubmed.ncbi.nlm.nih.gov/?term=%28%28%28

https://pubmed.ncbi.nlm.nih.gov/?term=%28%28%22Stomach+Neoplasms%22%5BMesh%5D%29+OR+%28%28%28

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9,10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9,10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11,12,13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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